THE FIBRATION QUESTION IN RANK THREE

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Abstract. We prove that the fibration question admits a positive answer in rank three whenever all non-split fibres are split by a common cubic extension of the base field.

1. Introduction

The Brauer–Manin obstruction provides a framework for studying the Hasse principle and weak approximation over number fields. For rationally connected varieties, the expected statement is expressed by the following conjecture of Colliot-Thélène [CT03, p. 174].

Conjecture 1.1 (Colliot-Thélène)**.** *Let k be a number field and let X be a smooth proper geometrically integral rationally connected k-variety. Then $X(k)$ is dense in $X(\mathbf{A}_k)^{\mathrm{Br}(X)}$.*

Fibrations over the projective line offer a way to approach this conjecture by passing from fibres over rational points to the total space. This leads to the following question.

Question 1.2 (The fibration question)**.** Let k be a number field, let X be a smooth proper geometrically integral k-variety, and let $f : X \to \mathbf{P}^1_k$ be a surjective morphism whose geometric generic fibre is rationally connected. Choose a dense open $U \subset \mathbf{P}^1_k$ over which the fibres are smooth and geometrically rationally connected. Suppose that there is a Hilbert subset $H_0 \subset \mathbf{P}^1_k$ such that $X_c(k)$ is dense in $X_c(\mathbf{A}_k)^{\mathrm{Br}(X_c)}$ for every $c \in U(k) \cap H_0$. Does it follow that $X(k)$ is dense in $X(\mathbf{A}_k)^{\mathrm{Br}(X)}$?

Descent theory and the fibration method offer complementary approaches to this question. Descent theory studies suitable torsors [CTS87], while the fibration method selects a rational parameter compatible with local conditions and fibrewise Brauer groups; the rank and the splitting fields govern this choice.

Harari developed the fibration method in the presence of the Brauer–Manin obstruction [Har94, Har97]. Subsequent developments encompass low-rank results, conditional theorems under Schinzel's hypothesis, and approaches via additive combinatorics; see [CSS98, CSk00, BMS14, HSW14, BS19]. Harpaz and Wittenberg first reduced the general fibration question to an approximation conjecture for auxiliary norm varieties [HW16, Proposition 9.9 and Theorem 9.17]. Harpaz, Wittenberg, and the author subsequently refined this framework and established a positive answer in rank at most two as well as in several rank-three cases [HWW22, Theorems 7.3 and 7.4]. In rank four, Xu, Zhu, and the author provide a

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positive answer when each non-split fibre over m is split by a quadratic extension of $k(m)$, using strong approximation for affine cones over intersections of two quadrics [WXZ25, Theorems 1.4 and 1.6(iii)].

The interplay with descent theory is illustrated by smooth proper models of the norm equation

$$P(t) = \mathrm{N}_{K/k}(z).$$

Using descent theory, Colliot-Thélène, Sansuc, and Swinnerton-Dyer proved that weak approximation with the Brauer–Manin obstruction holds on Châtelet surfaces, where $[K : k] = 2$ and P is separable of degree three or four [CTSS87a, CTSS87b]. Further results for norm equations, obtained by combining descent theory with geometric and analytic methods, can be found in [CTS89, CHS03, HBS02, SJ13, BM17, BH12, DSW15].

The norm-equation results recalled above are covered by the fibration theorems of [HW16, BS19, HWW22, WXZ25], apart from the cubic-extension case of [CTS89]. The present work recovers the weak-approximation statement with the Brauer–Manin obstruction in this remaining case through the fibration method over an arbitrary number field. More generally, it treats the case of a common cubic splitting field by means of a geometric construction of the auxiliary quadric fibrations.

Theorem 1.3. *Question 1.2 has a positive answer if* $\operatorname{rank}(f) \leq 3$ *and there is a cubic field extension* L/k *such that every fibre of* $f_L : X \otimes_k L \to \mathbf{P}^1_L$ *is split.*

Cases of rank at most two are already covered by [HWW22, Theorem 7.3], and the rank-three case of Theorem 1.3 is already known over $\mathbb{Q}$ by [HWW22, Theorem 7.4(i)–(ii)]. The argument below treats rank three over arbitrary number fields.

The theorem retains the fibrewise approximation hypothesis from the question. The cubic extension is fixed for all non-split fibres and need not be Galois. The proof actually establishes the stronger adelic assertion that, for every Hilbert subset $H \subset \mathbf{P}^1_k$,

$$\bigcup_{c \in U(k) \cap H} X_c(\mathbf{A}_k)^{\mathrm{Br}(X_c)} \quad \text{is dense in} \quad X(\mathbf{A}_k)^{\mathrm{Br}(X)}. \tag{1.1}$$

Applying this assertion with $H = H_0$ and then using the fibrewise hypothesis yields the conclusion of the question.

Our arithmetic input is strong approximation for affine norm cubics after deleting closed subsets of codimension at least two that contain the singular locus. Eliminating the base coordinates identifies the auxiliary norm variety with such an open subset, so its approximation property follows from the corresponding result for the cubic.

After normalizing the coefficients, the projection $\mathcal{C} \to \mathbf{A}^3_k$ yields unit fibres $q(a') = R(b)$, with q isotropic. Writing $q = x_1x_2 + dz^2$, the projection to x_1 has affine-space fibres away from zero. Two applications of [CTX13, Proposition 3.1] then establish strong approximation. For the outer projection, the fibres contained in the deleted set lie over a subset of codimension at least two in the base; removing its closure allows the criterion to apply directly.

Section 2 defines the norm cubic and identifies the auxiliary variety W with U_β. Section 3 describes its geometry and affine projection. Sections 4–5 prove strong approximation,

and Subsection 5.2 applies it to the fibration question. The examples in Section 6 follow [HWW22, Section 7.2.3].

1.1. **Terminology.** A variety over a field F is *split* if it contains a geometrically integral open subset over F (see [Sko96, Definition 0.1]). Thus a fibre X_m is split if it possesses a geometrically integral component of multiplicity one defined over its residue field $k(m)$. Let $M_0 \subset \mathbf{P}^1_k$ be the reduced locus of non-split fibres. Following [HWW22, Section 7.1], set

$$\operatorname{rank}(f) = \deg M_0 = \sum_{m \in M_0} [k(m) : k].$$

The common splitting hypothesis means that, for every closed point m and every field factor E of $k(m) \otimes_k L$, the fibre $X_m \otimes_{k(m)} E$ is split.

Write Ω_k for the places of k, Ω_∞ for its archimedean places, and $\mathbf{A}_k$ for its ring of adeles. Write k_ν for the completion at ν and $\mathcal{O}_\nu$ for its valuation ring when ν is finite. For a smooth proper k-variety V, we write $\operatorname{Br}(V)$ for its cohomological Brauer group. Its Brauer–Manin set is

$$V(\mathbf{A}_k)^{\operatorname{Br}(V)} = \left\{ (x_\nu) \in V(\mathbf{A}_k) : \sum_{\nu \in \Omega_k} \operatorname{inv}_\nu \alpha(x_\nu) = 0 \text{ for every } \alpha \in \operatorname{Br}(V) \right\}.$$

We say that V satisfies *weak approximation with the Brauer–Manin obstruction* if $V(k)$ is dense in this set. We use the usual adelic topology, including the archimedean local topologies.

For a possibly non-proper variety V, *strong approximation off a place* v_0 means that $V(k)$ is dense in $\operatorname{pr}^{v_0} V(\mathbf{A}_k)$, where pr^{v_0} omits the v_0-component. If $V(k_{v_0}) \neq \varnothing$, this projection is $V(\mathbf{A}_k^{v_0})$. Strong approximation is used below for the auxiliary norm varieties; the conclusion for the proper variety X is weak approximation.

The *affine cone* of a projective hypersurface defined by a homogeneous polynomial F is $\{F = 0\}$ in the corresponding affine space.

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2. The norm cubic and the auxiliary variety W

2.1. **The norm cubic and its boundary.** Let L/k be a cubic field extension, let $P \in k[t]$ be monic and separable of degree three, and set

$$R = k[t]/(P), \qquad \eta = t \bmod P, \qquad M = \operatorname{Spec}(R) \subset \mathbf{A}^1_k, \qquad E = L \otimes_k R.$$

We also write M for the set of closed points of M. For $m \in M$, let $a_m \in k(m)$ be the image of η. The algebra $L_m = L \otimes_k k(m)$ is a cubic étale $k(m)$-algebra, and $E \simeq \prod_{m \in M} L_m$ as an R-algebra.

Fix $\beta \in R^\times$. Let $\ell : R \to k$ be the linear map extracting the coefficient of η^2 in the basis $1, \eta, \eta^2$. Define

$$F_\beta(\tau) = \ell\big(\beta^{-1} \operatorname{N}_{E/R}(\tau)\big), \qquad C_\beta = \{F_\beta = 0\} \subset E \simeq \mathbf{A}^9_k,$$

and let $V_\beta \subset \mathbf{P}(E) \simeq \mathbf{P}^8_k$ be its projectivization. Thus C_β is the affine cone over the cubic hypersurface V_β.

To describe the boundary, extend scalars to $\overline{k}$ and choose splittings $L \otimes_k \overline{k} \simeq \overline{k}^3$ and $R \otimes_k \overline{k} \simeq \overline{k}^3$. Write the distinct roots of P as η_1, η_2, η_3, and denote the images of β by $\beta_1, \beta_2, \beta_3$. Use coordinates x_{ij} on $E \otimes_k \overline{k}$, with i indexing the embeddings of L and j indexing those of R. For $y \in R \otimes_k \overline{k}$, let y_j be its image under evaluation at η_j. Its unique polynomial representative $f_y(t)$ of degree at most two is

$$f_y(t) = \sum_{j=1}^{3} y_j \frac{P(t)}{(t-\eta_j)P'(\eta_j)}.$$

Since P is monic, each $P(t)/(t-\eta_j)$ is monic of degree two, so $\ell(y) = \sum_j y_j/P'(\eta_j)$. The j-th component of $\mathrm{N}_{E/R}(\tau)$ is $x_{1j}x_{2j}x_{3j}$. Applying this identity to $y = \beta^{-1}\,\mathrm{N}_{E/R}(\tau)$ therefore gives

$$F_\beta = \sum_{j=1}^{3} \gamma_j x_{1j}x_{2j}x_{3j}, \qquad \gamma_j = \frac{1}{\beta_j P'(\eta_j)} \neq 0. \tag{2.1}$$

The coefficients are nonzero because β is a unit and P is separable.

Let $D_\beta \subset C_\beta$ be the reduced closed subset whose geometric points have at least two of the nine coordinates x_{ij} equal to zero. This definition is independent of the chosen splittings: the Galois group permutes the nine absolute norm coordinates and their coordinate hyperplanes. Hence the indicated closed subset descends to k. Set

$$U_\beta = C_\beta \setminus D_\beta. \tag{2.2}$$

The boundary D_β will be precisely the deleted set in the comparison with the auxiliary norm variety.

Lemma 2.1. *The variety C_β is geometrically integral of dimension eight, and D_β is pure of codimension two in C_β. The singular locus of C_β has dimension three and is contained in D_β. In particular, U_β is smooth and geometrically integral. The projective cubic V_β is not a cone.*

Proof. Work over $\overline{k}$ and use (2.1). Regarded as a polynomial in x_{11}, its coefficient is $\gamma_1 x_{21}x_{31}$, whereas its constant term is $\gamma_2 x_{12}x_{22}x_{32} + \gamma_3 x_{13}x_{23}x_{33}$. These two polynomials are relatively prime in the polynomial ring in the other eight variables. Therefore the polynomial is primitive and linear in x_{11} over the corresponding fraction field, hence irreducible. This proves geometric integrality and dimension eight.

For the boundary, impose the vanishing of any two distinct coordinates. If they belong to the same column, the remaining equation is the sum of the two other triple products, a nonzero hypersurface equation in seven variables; every component has dimension six. If the coordinates belong to different columns, the equation reduces to the triple product in the third column, whose three components are coordinate six-spaces. Taking the finite union over all pairs proves that D_β is pure of dimension six.

The partial derivatives of F_β vanish exactly when at least two coordinates in each column vanish. Thus $\mathrm{Sing}(C_\beta)_{\overline{k}}$ is the union of the coordinate three-spaces obtained by allowing at most one coordinate in each column to vary. It has dimension three, hence codimension five in C_β, and is contained in D_β. Consequently the complement is smooth and geometrically integral. Finally, the nine partial derivatives are distinct nonzero quadratic monomials and are linearly independent. No nonzero constant direction annihilates F_β, so the projective hypersurface has no linear vertex. □

There are unique cubic polynomial functions λ, μ on C_β such that

$$\beta^{-1}\,\mathrm{N}_{E/R}(\tau) = \lambda(\tau) - \eta\mu(\tau).$$

Indeed, the coefficient of η^2 vanishes by definition, and $1, \eta$ are linearly independent. This identity will recover the two base coordinates of the auxiliary norm variety.

2.2. **The auxiliary norm variety.** With the notation above, write $\beta = (b_m)_{m\in M}$ under $R \simeq \prod_{m\in M} k(m)$, and set

$$F_m = \mathrm{Sing}(\mathrm{Res}_{L_m/k}\,\mathbf{A}^1 \setminus \mathrm{Res}_{L_m/k}\,\mathbf{G}_\mathrm{m}).$$

Define W by

$$\mathrm{N}_{L_m/k(m)}(z_m) = b_m(\lambda - a_m\mu) \qquad (m \in M) \tag{2.3}$$

inside

$$(\mathbf{A}^2_k \setminus \{0\}) \times \prod_{m\in M} (\mathrm{Res}_{L_m/k}\,\mathbf{A}^1 \setminus F_m).$$

It is smooth and geometrically integral, and its natural morphism to the projective line is

$$p : W \longrightarrow \mathbf{P}^1_k, \qquad (\lambda, \mu, (z_m)) \longmapsto [\lambda : \mu].$$

Geometrically, F_m is the locus where at least two absolute norm coordinates in the m-block vanish. Both exclusions in the definition of W are part of its integral model. When all L_m are fields, this is the auxiliary variety of [HWW22, Section 3.1]; the same construction applies to cubic étale algebras.

Lemma 2.2. *Projection to the norm coordinates induces a k-isomorphism*

$$W \simeq U_\beta = C_\beta \setminus D_\beta,$$

where C_β, D_β, and U_β are defined above.

Proof. Using $E \simeq \prod_m L_m$, the inverse is $\tau \mapsto (\lambda(\tau), \mu(\tau), \tau)$, with the cubic functions defined above. Over $\overline{k}$, two zero coordinates in one column lie in the corresponding F_m; two in different columns give $\lambda - \eta_j\mu = \lambda - \eta_h\mu = 0$, hence $\lambda = \mu = 0$. Conversely, either excluded condition forces at least two absolute coordinates to vanish. Thus the deleted locus is exactly D_β. □

3. Geometry of the norm cubic

3.1. **Normalizing arbitrary coefficients.** The two cubic functions λ, μ on C_β satisfy

$$\mathrm{N}_{E/R}(\tau) = \beta(\lambda - \eta\mu). \tag{3.1}$$

Lemma 3.1. *If C_β^{sm} has adelic points, there exists an invertible k-linear change of the nine norm coordinates identifying C_β with the normalized cubic C_1, for a possibly different generator of the same algebra R. This change identifies D_β with the corresponding normalized boundary. It may be constructed from any rational point $\tau_0 \in C_\beta(k) \cap E^\times$.*

Proof. By Lemma 2.1, the projective cubic $V_\beta \subset \mathbf{P}_k^8$ is geometrically integral and is not a cone. After splitting L and R, the points $\tau_{ij} = \delta_{ir}$ for $r = 1, 2, 3$ are three non-collinear singular points. They form a single Galois orbit indexed by the three embeddings of the cubic field L. Since the ambient space is $\mathbf{P}_k^8$, the main theorem of [CTS89] yields the Hasse principle and weak approximation for V_β^{sm}. Projecting the local smooth points and using local density of the open subset defined by $\mathrm{N}_{E/k}(\tau) \neq 0$ yields a rational point in this open subset. An affine lift is therefore a point

$$\tau_0 \in E^\times, \qquad \mathrm{N}_{E/R}(\tau_0) = \beta\delta, \qquad \delta = \lambda_0 - \eta\mu_0 \in R^\times.$$

Set $\zeta = \tau/\tau_0$. If $\mu_0 \neq 0$, define

$$\theta = \delta^{-1}, \qquad u = \mu/\mu_0, \qquad v = \lambda - \lambda_0\mu/\mu_0.$$

Then $\mathrm{N}_{E/R}(\zeta) = u + v\theta$, and since the three geometric values of θ are distinct, θ generates R. If $\mu_0 = 0$, take $\theta = \eta$, $u = \lambda/\lambda_0$, and $v = -\mu/\lambda_0$. In either case the change of the two base coordinates is invertible. Replacing v by $-v$ yields the normalized equation. Multiplication by τ_0^{-1} is an invertible linear change of the nine affine coordinates and rescales each absolute norm coordinate by a nonzero factor; hence it preserves the boundary. □

3.2. **The norm cubic and its polarization.** In the normalized coordinates of Lemma 3.1, write $\mathcal{C} = C_1$ and $D = D_1$ for the normalized cone and its boundary, and set

$$U = U_{\beta=1} = \mathcal{C} \setminus D.$$

The cubic functions recovered in Section 2 satisfy

$$\mathrm{N}_{E/R}(\tau) = \lambda - \eta\mu. \tag{3.2}$$

Translate the generator of R so as to arrange

$$P(t) = t^3 + \alpha_1 t + \alpha_0, \qquad \alpha_1, \alpha_0 \in k. \tag{3.3}$$

This preserves $k + k\eta$ and amounts to an invertible linear change of the affine coordinates, so it preserves the boundary open subset.

For $x \in L$, we use the notation

$$T(x) = \mathrm{Tr}_{L/k}(x), \quad q(x) = \sigma_2(x), \quad N(x) = \mathrm{N}_{L/k}(x), \quad x^\# = x^2 - T(x)x + q(x)1_L.$$

Here $\sigma_2(x)$ is the coefficient of z in the characteristic polynomial of the multiplication map $m_x : L \to L$:

$$\det(z - m_x) = z^3 - T(x)z^2 + q(x)z - N(x).$$

Regard q as a quadratic form on the three-dimensional k-vector space L.

Extending scalars to $\overline{k}$, write $L \otimes_k \overline{k} \simeq \overline{k}^3$. For $x = (x_1, x_2, x_3)$,

$$T(x) = x_1 + x_2 + x_3, \qquad q(x) = x_1x_2 + x_1x_3 + x_2x_3,$$
$$N(x) = x_1x_2x_3, \qquad x^{\#} = (x_2x_3, x_1x_3, x_1x_2).$$

Thus $x^{\#}$ is the adjugate of x, and

$$xx^{\#} = N(x)1_L, \qquad T(x^{\#}) = q(x), \qquad q(x^{\#}) = T(x)N(x). \tag{3.4}$$

The coordinate formulas also yield the polarization and scalar-shift identities

$$q(x+y) - q(x) - q(y) = T(x)T(y) - T(xy), \tag{3.5}$$

$$q(x + r1_L) = q(x) + 2rT(x) + 3r^2 \qquad (r \in k). \tag{3.6}$$

All of these are polynomial identities, so they descend to k and require no invertibility assumption on x.

Lemma 3.2. *The form q is nondegenerate and isotropic over k. In suitable k-linear coordinates,*

$$q(x, y, z) = xy + dz^2 \qquad (d \in k^{\times}).$$

Proof. The matrix of q in the split coordinates above has determinant $1/4$. Over L, the algebra $L \otimes_k L$ has the form $L \times E'$, with E' quadratic étale over L. The idempotent $(1, 0)$ is a nonzero isotropic vector for q_L. Springer's odd-degree theorem, applied to L/k, yields isotropy over k; see [Lam05, Chapter VII]. Witt decomposition yields the displayed form. □

Write uniquely

$$\tau = a + b\eta + c\eta^2, \qquad a, b, c \in L.$$

For a fixed coefficient $c_0 \in L$, write $U^{(c_0)} = U \cap \{c = c_0\}$; in particular, $U^{(1)} = U \cap \{c = 1_L\}$. Superscripts denote outer fibres, while the subscript β denotes the twisting coefficient. Let $\ell : R \to k$ extract the coefficient of η^2. Set

$$F(a, b, c) = \ell(\mathrm{N}_{E/R}(a + b\eta + c\eta^2)).$$

The coefficients of 1 and η determine λ, μ uniquely. Thus $\mathcal{C} = \{F = 0\} \subset \mathbf{A}^9_k$, and U is the open subset where at most one absolute norm coordinate vanishes.

Lemma 3.3. *One has*

$$F(a, b, c) = T(ca^{\#}) + T(a(b^{\#} - \alpha_1 c^{\#})) - \alpha_1 T(cb^{\#}) - \alpha_0 T(bc^{\#}) + \alpha_1^2 N(c). \tag{3.7}$$

For $c = 1$, the polynomial change of coordinates

$$a' = a - b^{\#} + \frac{q(b) - \alpha_1}{2} 1_L, \qquad a' \in L \tag{3.8}$$

yields

$$F(a, b, 1) = q(a') - R(b), \qquad R(b) = T(b)(N(b) + \alpha_0) - \frac{(q(b) - \alpha_1)^2}{4}. \tag{3.9}$$

Proof. In the split coordinates above, write

$$\prod_{i=1}^{3}(a_i + b_i u + c_i u^2) = \sum_{j=0}^{6} d_j u^j$$

for an indeterminate u, before reduction modulo P. The coefficients contributing to F are

$$d_2 = T(ca^{\#}) + T(ab^{\#}), \qquad d_4 = T(ac^{\#}) + T(cb^{\#}),$$
$$d_5 = T(bc^{\#}), \qquad d_6 = N(c).$$

The relation $\eta^3 = -\alpha_1\eta - \alpha_0$ yields

$$\eta^4 = -\alpha_1\eta^2 - \alpha_0\eta,$$
$$\eta^5 = -\alpha_0\eta^2 + \alpha_1^2\eta + \alpha_1\alpha_0,$$
$$\eta^6 = \alpha_1^2\eta^2 + 2\alpha_1\alpha_0\eta + \alpha_0^2.$$

Since $\ell(1) = \ell(\eta) = \ell(\eta^3) = 0$, we obtain

$$F = d_2 - \alpha_1 d_4 - \alpha_0 d_5 + \alpha_1^2 d_6,$$

which is (3.7).

For $c = 1_L$, (3.7) gives

$$F(a, b, 1) = q(a) + T(ab^{\#}) - \alpha_1 T(a) - \alpha_1 q(b) - \alpha_0 T(b) + \alpha_1^2. \tag{3.10}$$

Put $s = q(b)$ and $h = -b^{\#} + (s - \alpha_1)1_L/2$, so that $a' = a + h$. The identities (3.4)–(3.6) give

$$q(a') = q(a) + T(ab^{\#}) - \alpha_1 T(a) + q(h),$$
$$q(h) = T(b)N(b) - s(s - \alpha_1) + \frac{3}{4}(s - \alpha_1)^2.$$

Substitution into (3.10) gives $F(a, b, 1) = q(a') - R(b)$. The change has polynomial inverse $a = a' + b^{\#} - (q(b) - \alpha_1)1_L/2$. □

3.3. **The affine projection.** The outer fibration is the morphism

$$\pi : \mathcal{C} \longrightarrow \mathrm{Res}_{L/k}\, \mathbf{A}^1_L \simeq \mathbf{A}^3_k, \qquad (a, b, c) \longmapsto c. \tag{3.11}$$

For $N(c) \neq 0$, dividing (a, b, c) by c and using (3.8) identifies the fibre with

$$Y = \{q(a') = R(b)\} \subset \mathbf{A}^3_k \times \mathbf{A}^3_k.$$

In the hyperbolic coordinates $q = x_1x_2 + dz^2$, the projection $Y \to \mathbf{A}^1_k$ to x_1 has fibres $\mathbf{A}^4_k$ for $x_1 \neq 0$, obtained by solving for x_2. These two affine projections provide the fibrations needed below.

4. Arithmetic on the quadric fibres

We prove strong approximation for isotropic quadratic equations after deleting subsets of codimension at least two, and then apply it to the unit outer fibres.

4.1. **An affine-space fibration.**

Lemma 4.1. *Let k be a number field and let $d \geq 2$. If $E \subset \mathbf{A}^d_k$ is closed of codimension at least two, then $\mathbf{A}^d_k \setminus E$ satisfies strong approximation off any place v_0 of k.*

For this statement about open subsets of affine space, see [Wei21, Lemma 2.1]. Strong approximation on $\mathbf{A}^1_k$ will also be used.

Theorem 4.2. *Let q be a nondegenerate isotropic quadratic form over a number field k in $m \geq 3$ variables. Let $p \in k[t_1, \ldots, t_n]$ be nonconstant. Suppose that either $m \geq 4$, or $m = 3$ and*

$$p \neq ch^2 \qquad (c \in k^\times,\ h \in k[t_1, \ldots, t_n]).$$

Let V be the affine hypersurface

$$V = \{q(x_1, \ldots, x_m) = p(t_1, \ldots, t_n)\}.$$

For every closed subset $Z \subset V^{\mathrm{sm}}$ of codimension at least two, the complement $V^{\mathrm{sm}} \setminus Z$ satisfies strong approximation off any place v_0 of k.

Proof. Write $q = x_1x_2 + q'(x_3, \ldots, x_m)$, and let $\pi : V \to \mathbf{A}^1_k$ be the projection to x_1. The defining polynomial is primitive and linear in x_2, so V is geometrically integral. Set $d = m + n - 2$. For $a \neq 0$,

$$V_a \simeq \mathbf{A}^d_k, \qquad x_2 = \frac{p(t) - q'(x_3, \ldots, x_m)}{a}.$$

The zero fibre is

$$V_0 = \mathbf{A}^1_{x_2} \times \{q'(x_3, \ldots, x_m) = p(t)\}.$$

It is geometrically integral. For $m = 3$, a factorization of p as a square up to a constant over $\overline{k}$ would give even multiplicities for all geometric factors, hence $p = ch^2$ over k, contrary to the hypothesis. For $m \geq 4$, its second factor has, over $\overline{k}$, the irreducible equation $uv = p(t) - q''(z)$.

Set $X = V^{\mathrm{sm}} \setminus Z$. Since

$$\mathrm{Sing}(V) \subset \{x_1 = \cdots = x_m = 0,\ p(t) = 0\},$$

its dimension is at most $n - 1$, so it has codimension at least m. The closure of Z in V also has codimension at least two. Thus every geometric fibre of $X \to \mathbf{A}^1$ is nonempty and integral. There exists a nonempty open $G \subset \mathbf{G}_{\mathrm{m}}$ such that $Z \cap V_a$ has codimension at least two for $a \in G$. Above G the morphism is smooth, and its rational fibres satisfy strong approximation off v_0 by Lemma 4.1. Moreover, $\pi^{-1}(G)(k_{v_0}) \to G(k_{v_0})$ is surjective, since these fibres are nonempty open subsets of affine space. Strong approximation on $\mathbf{A}^1$ and the fibration criterion [CTX13, Proposition 3.1] therefore yield strong approximation for X off v_0. □

4.2. **The unit outer fibre.** Return to the normalized norm cone $\mathcal{C}$ and its open subset $U = U_{\beta=1}$ from (2.2). Fix a closed subset $Z \subset \mathcal{C}$ of codimension at least two. Under the quadratic change (3.8), the fibre $U^{(1)} = U \cap \{c = 1_L\}$ is identified with an open subset of

$$Y = \{q(a') = R(b)\} \subset \mathbf{A}^3_{a'} \times \mathbf{A}^3_b.$$

The quadratic form q is nondegenerate and isotropic by Lemma 3.2. On the scalar line $b = s1_L$,

$$R_0(s) := R(s1_L) = \frac{3s^4 + 6\alpha_1 s^2 + 12\alpha_0 s - \alpha_1^2}{4}. \tag{4.1}$$

Its derivative is $R_0'(s) = 3P(s)$. At a root r of $P(s) = s^3 + \alpha_1 s + \alpha_0$, direct substitution yields

$$4R_0(r) = -(3r^2 + \alpha_1)^2 = -P'(r)^2.$$

Since P is separable, R_0 has no multiple root. In particular $R(b)$ is nonconstant and cannot be a constant times a square, even over $\overline{k}$: its restriction to this scalar line would then be a square up to a constant, contrary to (4.1).

Lemma 4.3. *The complement $D^{(1)} = Y \setminus U^{(1)}$ is closed of codimension at least two in Y and contains* $\mathrm{Sing}(Y)$. *The open subset $U^{(1)}$ is smooth and geometrically integral, and $U^{(1)}(k)$ is Zariski dense.*

Proof. Recall that $U = \mathcal{C} \setminus D$, where $D = D_{\beta=1}$ is the closed subset whose geometric points have at least two absolute norm coordinates equal to zero. Division by c yields the following isomorphisms:

$$\mathcal{C} \cap \{N(c) \neq 0\} \simeq Y \times \mathrm{Res}_{L/k}\,\mathbf{G}_{\mathrm{m}}, \qquad U \cap \{N(c) \neq 0\} \simeq U^{(1)} \times \mathrm{Res}_{L/k}\,\mathbf{G}_{\mathrm{m}}. \tag{4.2}$$

Indeed, divide (a, b, c) by c, and then apply (3.8) on the fibre $c = 1_L$. This operation preserves the boundary: after splitting the algebras, it multiplies every absolute norm coordinate by a nonzero factor. Consequently the boundary $D \cap \{N(c) \neq 0\}$ on the first open subset in (4.2) corresponds to $D^{(1)} \times \mathrm{Res}_{L/k}\,\mathbf{G}_{\mathrm{m}}$. By Lemma 2.1, this boundary has dimension at most six, so $\dim D^{(1)} \leq 3$, whereas $\dim Y = 5$. Since U is smooth and geometrically integral, the second product in (4.2) proves the same properties for $U^{(1)}$. As $U^{(1)}$ is a smooth open subset of Y, every singular point of Y lies in its complement; hence $\mathrm{Sing}(Y) \subset Y \setminus U^{(1)} = D^{(1)}$. Finally, in the hyperbolic coordinates for q, the open locus $x_1 \neq 0$ in Y is $\mathbf{G}_{\mathrm{m}} \times \mathbf{A}^4$. Its rational points are Zariski dense, and removing $D^{(1)}$ preserves this density. □

Proposition 4.4. *Let $c \in L^\times$, and suppose $Z \cap U^{(c)}$ has codimension at least two in $U^{(c)}$. Then $U^{(c)} \setminus Z$ satisfies strong approximation off every place of k.*

Proof. Multiplication of τ by c^{-1} identifies $U^{(c)}$ with $U^{(1)}$. Let $Z^{(1)} \subset U^{(1)}$ be the image of $Z \cap U^{(c)}$. Its closure in Y has codimension at least two, as does $D^{(1)}$ by Lemma 4.3. Moreover,

$$U^{(1)} \setminus Z^{(1)} = Y^{\mathrm{sm}} \setminus \big((D^{(1)} \cup \overline{Z^{(1)}}^{\,Y}) \cap Y^{\mathrm{sm}}\big).$$

Theorem 4.2, applied to $q(a') = R(b)$, therefore proves the assertion. □

5. Strong approximation and the fibration question

5.1. Strong approximation for the norm cubic.

Theorem 5.1. *Let k be a number field and let C_β be the norm cubic of Section 2. For every closed subset $Z \subset C_\beta$ of codimension at least two containing* $\mathrm{Sing}(C_\beta)$*, the complement $C_\beta \setminus Z$ satisfies strong approximation off every place of k.*

Lemma 5.2. *Set*

$$\mathcal{B}_0 = \mathbf{A}^3_k \setminus \mathrm{Sing}\{N(c) = 0\}.$$

Every geometric fibre $\mathcal{C}_c$ of (3.11) over $\mathcal{B}_0$ is integral of dimension five.

Proof. If $N(c) \neq 0$, the fibre is isomorphic to Y, which is geometrically integral by Theorem 4.2 and its proof. Otherwise, after splitting L and renumbering, write $c = (0, c_2, c_3)$ with $c_2c_3 \neq 0$. Formula (3.7) becomes

$$F = Ua_2 + Va_3 + H,$$

where

$$U = a_1c_3 + b_1b_3, \qquad V = a_1c_2 + b_1b_2,$$
$$H = a_1b_2b_3 - \alpha_1a_1c_2c_3 - \alpha_1b_1(c_2b_3 + c_3b_2) - \alpha_0b_1c_2c_3.$$

The polynomials U, V are coprime: U is irreducible, and substituting $a_1 = -b_1b_3/c_3$ in V yields $b_1(b_2 - c_2b_3/c_3) \neq 0$. Thus F is primitive over $\overline{k}[a_1, b_1, b_2, b_3]$ and linear in (a_2, a_3), hence irreducible. This proves the assertion. □

Proof of Theorem 5.1. Assume that the adelic set of $C_\beta \setminus Z$ is nonempty. After normalizing by Lemma 3.1, we may assume that $\beta = 1$. Set $E_0 = Z \cup D$, where $D = \mathcal{C} \setminus U$. Then $\dim E_0 \leq 6$. Let $H_0 \subset \mathcal{B}_0$ be the locus where the geometric fibre $\mathcal{C}_c$ is contained in E_0. It is constructible by Chevalley's theorem, as the complement of the image of $\mathcal{C} \setminus E_0$. By Lemma 5.2, its fibres inside E_0 all have dimension five, so

$$\dim H_0 + 5 \leq \dim E_0 \leq 6.$$

Its closure H in $\mathbf{A}^3_k$ therefore has dimension at most one. Consequently

$$\mathcal{B} = \mathcal{B}_0 \setminus H$$

is the complement of a closed subset of codimension at least two in $\mathbf{A}^3_k$, and satisfies strong approximation off v_0 by Lemma 4.1.

Let $X' = (U \setminus Z) \cap \pi^{-1}(\mathcal{B})$. Every geometric fibre of $X' \to \mathcal{B}$ is a nonempty open subset of the corresponding integral fibre of Lemma 5.2. The fibre dimension theorem yields a nonempty open $G \subset \mathcal{B} \cap \{N(c) \neq 0\}$ such that $Z \cap U^{(c)}$ has codimension at least two for all $c \in G$. The morphism is smooth over G by the product isomorphism (4.2). Its rational fibres there satisfy strong approximation by Proposition 4.4. For every $c \in G(k_{v_0})$, division by c identifies the fibre with a nonempty open subset of the rational variety Y, which contains $\mathbf{G}_\mathrm{m} \times \mathbf{A}^4$ as a dense open subset. Hence the fibre has a k_{v_0}-point.

All hypotheses of [CTX13, Proposition 3.1] now hold for $X' \to \mathcal{B}$. It follows that X' satisfies strong approximation off v_0; the criterion supplies the integral conditions, including

at the new primes dividing $N(c)$. Finally, $\mathcal{C} \setminus Z$ is smooth, since Z contains $\mathrm{Sing}(\mathcal{C})$, and is geometrically integral. The variety X' is a nonempty open subset of it, hence dense. Smoothness gives density of $X'(k_v)$ in $(\mathcal{C} \setminus Z)(k_v)$ at every place v, and a suitable integral model of X' has integral points at almost every finite place. Thus [CTX13, Proposition 2.3] transfers strong approximation off v_0 from X' to $\mathcal{C} \setminus Z$. □

Corollary 5.3. *For every $\beta \in R^\times$, the variety*

$$U_\beta = C_\beta \setminus D_\beta$$

satisfies strong approximation off every place of k.

Proof. By Lemma 2.1, D_β has codimension two in C_β and contains $\mathrm{Sing}(C_\beta)$. Apply Theorem 5.1 with $Z = D_\beta$. □

5.2. **Proof of the main theorem.** We now relate the splitting fields in the fibration to the auxiliary norm fields.

Proof of Theorem 1.3. Choose a coordinate on the base so that infinity lies in U. For $m \in M_0$, set $F = k(m)$ and $L_m = F \otimes_k L$. If L_m were not a field, a defining cubic polynomial for L/k would have a root in F. Since $[F : k] \leq 3$, this would force $F \cong_k L$. The resulting F-factor of $F \otimes_k L$ would make X_m split, a contradiction. Thus L_m/F is cubic. The reduced image in X_m of a geometrically integral multiplicity-one component over L_m is a multiplicity-one component whose constant field embeds in L_m. This field cannot be F, since X_m is non-split; hence it is L_m. At split fibres we choose $L_m = k(m)$.

For arbitrary auxiliary data as in [HWW22, Section 3.6], the nontrivial norm fields occur at $M \cap M_0$, of total degree at most three. If this degree is at most two, [HWW22, Corollary 6.2(i)] proves F_+. If it is three, eliminate the trivial norm equations using [HWW22, Remark 3.2]. By Lemma 2.2, the resulting variety is U_β. Corollary 5.3 gives strong approximation off every place; hence [HWW22, Proposition 6.1] proves F_+ for the full parameters, retaining the auxiliary abelian extensions at split points.

The rational connectedness of the geometric generic fibre supplies the remaining hypotheses of [HWW22, Corollary 4.7 and Remarks 4.8(i)–(ii)], which therefore yields (1.1). Taking $H = H_0$ and applying the fibrewise approximation hypothesis yields the required density of $X(k)$ in $X(\mathbf{A}_k)^{\mathrm{Br}(X)}$. □

6. Norms from a ramified cubic cover

Section 7.2.3 of [HWW22] constructs fibrations from a finite cover of the base whose ramification indices have gcd two. Here we give an analogous construction with gcd three. It yields examples in which the norm torus itself varies with the parameter.

Let $P \in k[t]$ be separable of degree three, let $M = \{P = 0\}$, and let $\pi : C \to \mathbf{P}^1_k$ be the finite map obtained by normalizing and completing

$$u^3 = P(t).$$

The curve C is smooth, projective, and geometrically integral. The map π has degree three, is étale off M, and is totally ramified of index three at each $m \in M$, with residue field $k(m)$. At infinity, writing $s = t^{-1}$ and $u' = su$ yields

$$(u')^3 = s^3 P(s^{-1}).$$

The right-hand side is a unit at $s = 0$, so infinity is unramified. These assertions also follow from the Kummer description after adjoining the cube roots of unity; the ramification statements descend to k.

For $\beta \in k^\times$, define

$$X^0 = \{\mathrm{N}_{C/\mathbf{P}^1_k}(z) = \beta\} \subset \mathrm{Res}_{C/\mathbf{P}^1_k} \mathbf{A}^1_C, \qquad f^0 : X^0 \longrightarrow \mathbf{P}^1_k.$$

This relative Weil restriction is taken along the finite locally free map π.

Lemma 6.1. *The morphism f^0 is smooth. Its fibres off M are geometrically integral torsors under norm tori. For $m \in M$, writing $F = k(m)$, its fibre is isomorphic to*

$$\mathrm{Spec}(F[x]/(x^3 - \beta)) \times \mathbf{A}^2_F.$$

For every field factor E of $F \otimes_k L$, where $L = k(\sqrt[3]{\beta})$, its base change to E has a multiplicity-one component isomorphic to $\mathbf{A}^2_E$ and is therefore split.

Proof. Since $\beta \neq 0$, every point of the norm fibre is a unit in the corresponding finite algebra. For a unit z, the differential of the norm is

$$d\,\mathrm{N}_z(h) = \mathrm{N}(z)\,\mathrm{Tr}(z^{-1}h).$$

The trace is surjective: the trace of 1 is 3, which is invertible over k. The unit Weil restriction is smooth over the base, and this differential shows that the norm map, and hence its fibre over β, is smooth over the base. Off M, the algebra is étale of rank three; over an algebraic closure the norm equation is $z_1 z_2 z_3 = \beta$, a torsor under a two-dimensional torus. At m, the algebra is $F[v]/(v^3)$. The norm of $x_0 + x_1 v + x_2 v^2$ is x_0^3. In each field factor E, the image of $\sqrt[3]{\beta}$ is a simple root of $x^3 - \beta$, yielding the asserted component. □

Corollary 6.2. *For every $\beta \in k^\times$, every smooth proper model X of X^0 satisfies weak approximation with the Brauer–Manin obstruction. If the induced rational map to $\mathbf{P}^1_k$ extends to a morphism $f : X \to \mathbf{P}^1_k$, then* $\mathrm{rank}(f) \leq 3$*, and every fibre of f_L is split for $L = k(\sqrt[3]{\beta})$. If β is a cube in k, all fibres of f are split.*

Proof. The smooth variety X^0 is geometrically integral because it is flat over the base and has geometrically integral generic fibre. By invariance of fibre splitting on smooth proper models over the base [Sko96, Corollary 1.2], we may compactify X^0 over $\mathbf{P}^1_k$ and resolve the boundary, retaining X^0. Lemma 6.1 shows that fibres outside M are split, so

$$\mathrm{rank}(f) \leq \sum_{m \in M} [k(m) : k] = 3.$$

It also shows that every fibre becomes split over L: the geometrically integral multiplicity-one components supplied by the lemma persist in the proper model. If β is a cube in k, the same lemma shows that all fibres are already split over their residue fields.

Over a dense open subset, the proper fibres compactify torsors under $\mathrm{Res}^1_{C_c/k}\mathbf{G}_\mathrm{m}$, where C_c is the étale fibre of π at c. The geometric generic fibre of f is rational, and its smooth fibres over k-rational points satisfy weak approximation with the Brauer–Manin obstruction by [Sko01, Theorem 6.3.1]. If β is not a cube in k, then L/k is cubic and Theorem 1.3 applies. If β is a cube in k, all fibres are split, so [HWW22, Theorem 7.3] applies. Finally, use birational invariance for an arbitrary smooth proper model. □

Remark 6.3. In Corollary 6.2, if β is not a cube in $k(m)$ for any $m \in M$, then $\mathrm{rank}(f) = 3$. In particular, if the three branch points are k-rational, it suffices that $\beta \notin k^{\times 3}$.

Proof. Work on a model containing X^0. Fix $m \in M$ and put $F = k(m)$. If X_m had a geometrically integral multiplicity-one component D, a local parameter of m would have valuation one at D. The cubic cover would therefore be totally ramified after completion at the generic point of D, with residue field $F(D)$. Since $\mathrm{N}(z) = \beta$ is a unit, z is a unit, and reducing the norm equation gives $\overline{z}^3 = \beta$ in $F(D)$. But F is algebraically closed in $F(D)$, so β would be a cube in F, a contradiction. Thus every fibre over M is non-split. All other fibres are split, hence $\mathrm{rank}(f) = \sum_{m\in M}[k(m) : k] = 3$. □

Example 6.4. Take $k = \mathbb{Q}$, $P(t) = t^3 - 2$, and $\beta = 5$. On the affine chart the norm equation is

$$x_0^3 + (t^3-2)x_1^3 + (t^3-2)^2x_2^3 - 3(t^3-2)x_0x_1x_2 = 5. \tag{6.1}$$

A smooth proper model is a threefold satisfying weak approximation with the Brauer–Manin obstruction. Put $F = \mathbb{Q}(\sqrt[3]{2})$. Since $5 \notin F^{\times 3}$, Remark 6.3 gives rank three, with the unique non-split fibre above $t^3 - 2 = 0$. Every fibre is split by the non-Galois field $\mathbb{Q}(\sqrt[3]{5})$.

Example 6.5. For a cover ramified at the three rational points $0, 1, \infty$, take the smooth projective normalization of $u^3 = t(t-1)$ and choose $\beta \in k^\times \setminus k^{\times 3}$. This cover is étale away from $0, 1, \infty$ and totally ramified of degree three at all three points. At infinity, $s = t^{-1}$ and $u' = su$ give $(u')^3 = s(1-s)$, so the same local calculation applies. The corresponding relative norm equation $\mathrm{N}_{C/\mathbf{P}^1_k}(z) = \beta$ therefore has a smooth proper model of rank three by Remark 6.3, with common splitting field $k(\sqrt[3]{\beta})$, and satisfies weak approximation with the Brauer–Manin obstruction.

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